

Mixture viscosity measurement --- Solution

W26 (2026) — English translation

A1 Express the mole fraction of water n_w in solution in terms of the mass fraction of glycerol w_g , μ_w and μ_g . **0.20**

The amount of a substance in a mixture is defined as:

$$\nu = \frac{m}{\mu},$$

where m is the mass of the substance, and μ is its molar mass. Let's write the formula for the molar concentration of water:

$$n_w = 1 - n_g = 1 - \frac{\nu_g}{\nu_g + \nu_w},$$

from here:

$$\nu = \frac{m}{\mu},$$

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Let's write the formula for the molar concentration of water:

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from here:

Answer:

$$n_w = 1 - \frac{w_g \mu_w}{\mu_g - w_g(\mu_g - \mu_w)}$$

A2 Using the equation (1), convert the values of T_g to the values of w_g . **1.50**

Using the table method or the method of simple iterations with $a = 0.9$, we get:

Answer: T_g, K $w_g \eta, Pa \cdot c$ 195.2 1.000 0.906 194.8 0.996 0.839 194.6 0.994 0.767 193.9 0.987 0.655 193.1 0.979 0.558 191.8 0.965 0.479 190.8 0.954 0.373 188.6 0.930 0.202 183.4 0.869 0.0674 173.0 0.728 0.0278 170.0 0.682 0.0134 154.6 0.384 0.0026 149.3 0.252 0.0017 143.8 0.100 0.0012 140.3 0 0.0009

A3 Propose a linearization that describes a system with a small mole fraction of water, which can be used to determine the parameters ξ and k . Plot the linearized relationship. Determine the parameters ξ and k using the resulting graph. 0.60

Let's recalculate the points into the dependence $\ln \frac{\eta}{1 - \Pi q \cdot c} (n_w)$.

$T_g, Kw_g \eta, Pa \cdot c n_w \ln \eta$ 195.2 1.000 0.906 0 -0.099 194.8 0.996 0.839 0.020 -0.176 194.6 0.994 0.767 0.030
-0.265 193.9 0.987 0.655 0.063 -0.423 193.1 0.979 0.558 0.099 -0.583 191.8 0.965 0.479 0.156 -0.736 190.8 0.954

0.373 0.198 -0.986 188.6 0.930 0.202 0.278 -1.600 183.4 0.869 0.0674 173.0 0.728 0.0278 170.0 0.682 0.0134 154.6
0.384 0.0026 149.3 0.252 0.0017 143.8 0.100 0.0012 140.3 0 0.0009

Let's plot the resulting dependence in the range $n_w \in [0, 0.2]$.

$$\ln \frac{\eta}{1 \Pi a \cdot c}(n_w)$$

Approximating the dependence of the straight line, we obtain:

$$k = 4.31,$$

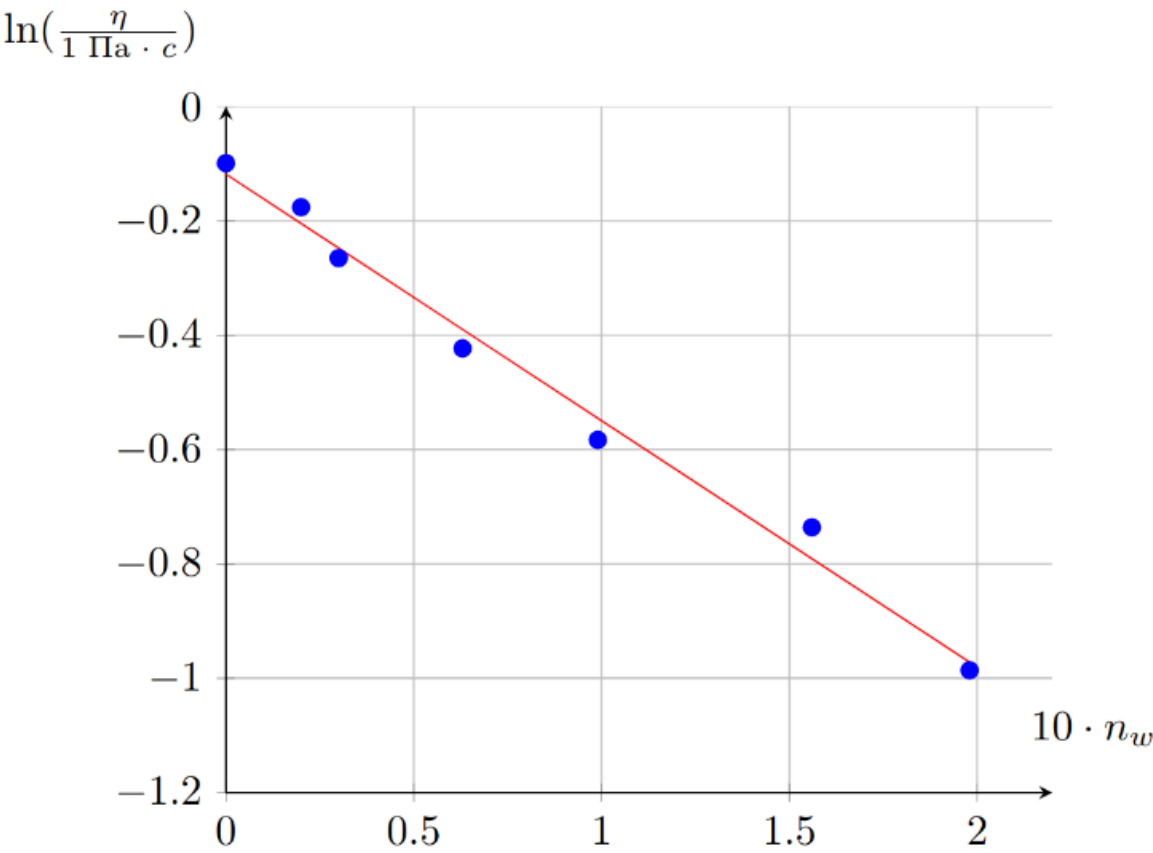
$$\ln \frac{\xi}{1 \Pi a \cdot c} = -0.12,$$

from which it follows:

Answer:

$$k = 4.31$$

$$\xi = 0.887 \Pi a \cdot c$$



A4 Knowing the parameters ξ and k , propose a linearization that allows you to determine the values of the parameters a and b . Plot the linearized relationship. Using the straight section, determine a and b . 0.50

Let's continue recalculating points:

$T_g, K w_g \eta, \Pi a \cdot c n_w \ln \left| \frac{\xi \exp(-k n_w)}{\eta} - 1 \right|$ 195.2 1.000 0.906 0 -3.92 194.8 0.996 0.839 0.020 -3.54 194.6 0.994
 0.767 0.030 -4.12 193.9 0.987 0.655 0.063 -3.44 193.1 0.979 0.558 0.099 -3.28 191.8 0.965 0.479 0.156 -2.91 190.8
 0.954 0.373 0.198 -4.35 188.6 0.930 0.202 0.278 -1.12 183.4 0.869 0.0674 0.435 0.02 173.0 0.728 0.0278 0.656
 -0.12 170.0 0.682 0.0134 0.704 0.78 154.6 0.384 0.0026 0.891 1.85 149.3 0.252 0.0017 0.938 2.10 143.8 0.100
 0.0012 0.979 2.29 140.3 0 0.0009 1 2.50

Let's plot the resulting points on the graph:

$$\ln \left| \frac{\xi \exp(-k n_w)}{\eta} - 1 \right| (n_w)$$

In the range $n_w \in [0, 0.2]$, the expression under the logarithm has a different sign and is near zero, so the function in this region behaves non-monotonic. To draw a straight line, we take into account only points in the range $n_w \in [0.7, 1]$.

$$\ln \left| \frac{\xi \exp(-k n_w)}{\eta} - 1 \right| (n_w) \text{ in the range } n_w \in [0.7, 1] \text{ (not required from participants)}$$

For $n_w > 0.2$, $\frac{\xi \exp(-k n_w)}{\eta} > 1$ holds, so $a > 0$. Having drawn the approximating straight line, we find:

$$b = 5.66,$$

$$\ln a = -3.21,$$

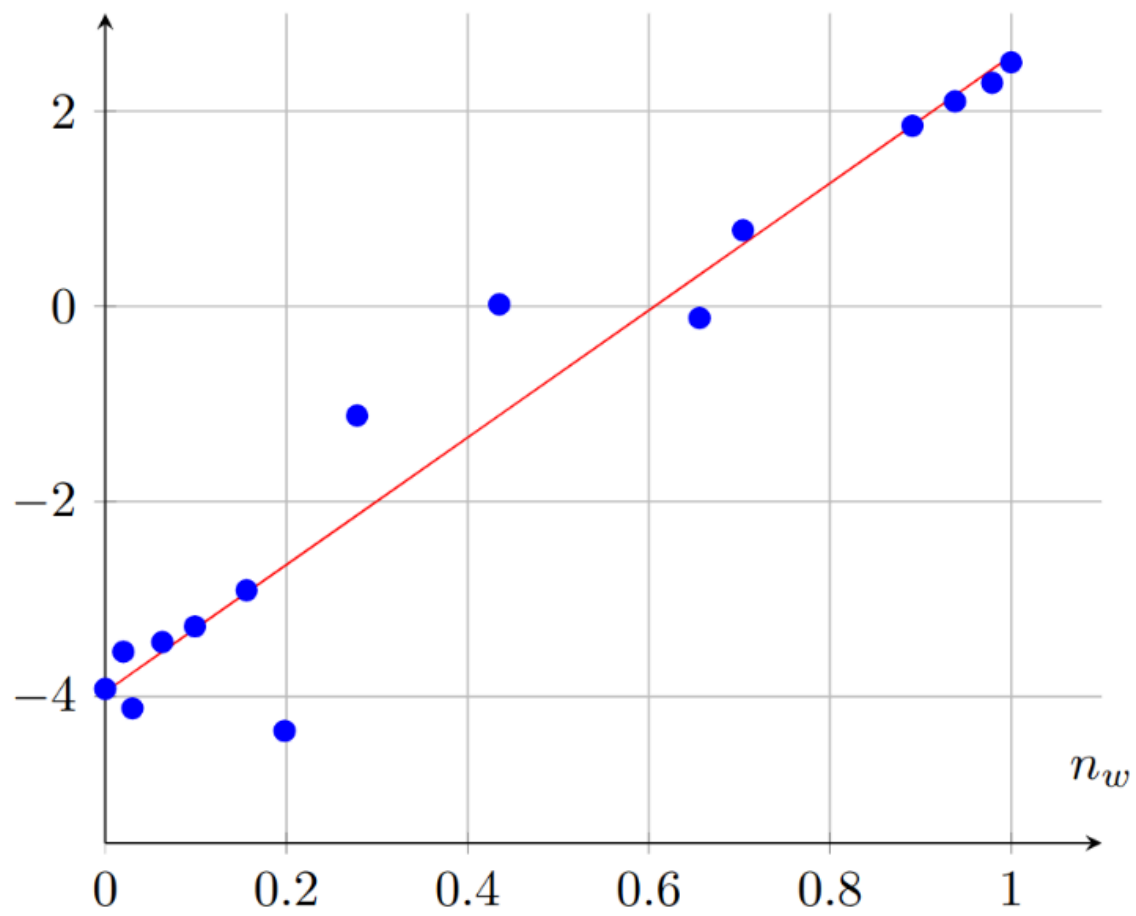
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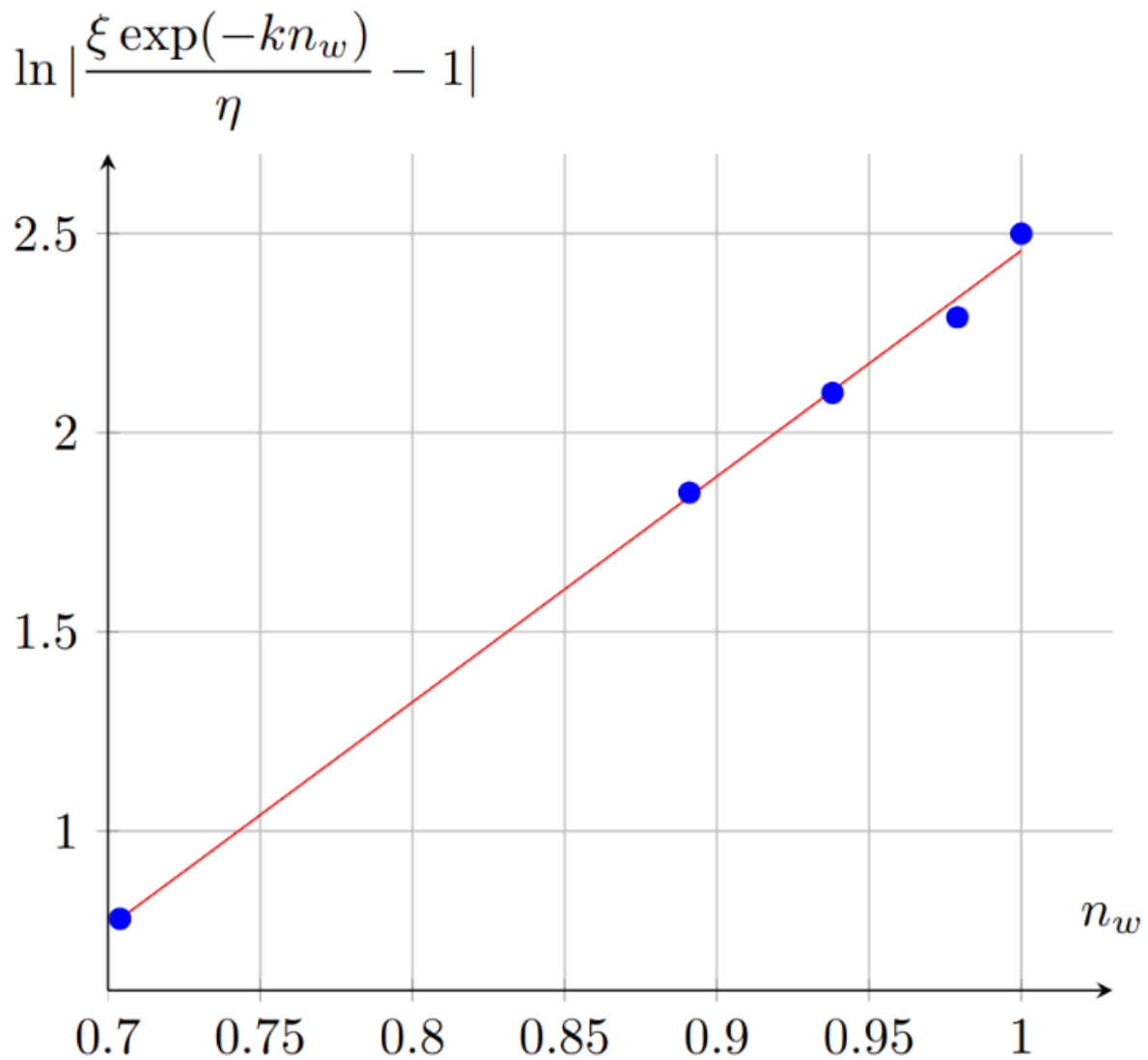
Answer:

$$a = 0.040$$

$$b = 5.66$$

$$\ln \left| \frac{\xi \exp(-kn_w)}{\eta} - 1 \right|$$





A5 Draw a graph of the theoretical and experimental dependence $\eta(n_w)$ on one piece of graph paper. 0.70
Indicate the range of applicability of the resulting equation.

Let's calculate the values of the theoretical dependence $\eta(n_w)$:

T_g, K	w_g	η	$\Pi a \cdot c$	n_w	$\ln \eta$	$\ln \left \frac{\xi \exp(-kn_w)}{\eta} - 1 \right $	η_{meop}	$\Pi a \cdot c$
195.2	1.000	0.906	0	-0.099	-3.92	0.853	194.8	0.996
0.839	0.020	-0.176	-3.54	0.779	194.6	0.994	0.767	0.030
-0.265	-4.12	0.744	193.9	0.987	0.655	0.063	-0.423	-3.44
0.640	193.1	0.979	0.558	0.099	-0.583	-3.28	0.541	191.8
0.965	0.479	0.156	-0.736	-2.91	0.413	190.8	0.954	0.373
0.198	-0.986	-4.35	0.337	188.6	0.930	0.202	0.278	-1.600
-1.12	0.224	183.4	0.869	0.0674	0.435	0.02	0.0926	173.0
0.728	0.0278	0.656	-0.12	0.0199	170.0	0.682	0.0134	0.704
0.78	0.0135	154.6	0.384	0.0026	0.891	1.85	0.0026	149.3
0.252	0.0017	0.938	2.10	0.0017	143.8	0.100	0.0012	0.979
2.29	0.0012	140.3	0	0.0009	1	2.50	0.0010	

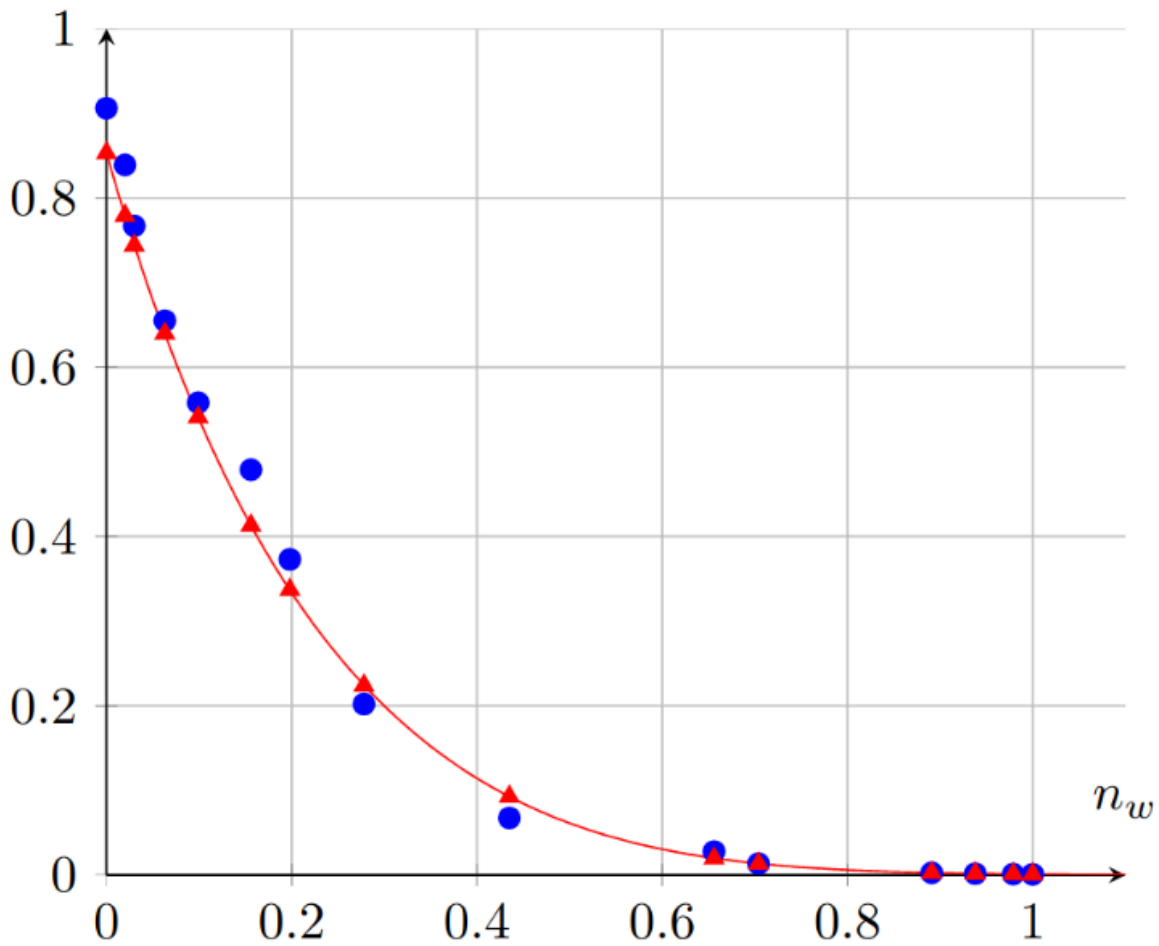
Let's build a graph of the obtained dependencies:

$\eta(n_w)$

Having drawn smoothing curves, we conclude:

Answer: The equation is applicable over the entire range ($n_w \in [0, 1]$).

$\eta, \text{Па} \cdot \text{с}$



B1 Using the equation (2), determine the glass transition temperature of the solution.

0.40

Let's express w_g :

$$w_g = \frac{m_g}{m_g + m_w} = \frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w},$$

then:

Answer:

$$T_g = (140.3 + 35.272 \frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} - 3.879 (\frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w})^2 + 23.467 (\frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w})^3) K \approx 162.8 K$$

B2 Propose a linearization to determine the values of the parameters D and T_0 . By plotting a linearized relationship or using the least squares method (OLS), determine the parameter values.

1.50

The dependence $\frac{1}{\ln \frac{\eta}{\eta_0}}(T)$ is linear with slope $k = \frac{1}{DT_0}$ and intercept term $b = -\frac{1}{D}$. Let's recalculate the point using this formula and plot the linearized dependence:

$T, ^\circ C$ $\eta, Pa \cdot s$ T, K $\frac{1}{\ln \frac{\eta}{\eta_0}}$ 1 0.0201 274 0.1149 4 0.0173 277 0.1169 11 0.0128 284 0.1212 15 0.0104 288 0.1243
 35 0.0046 308 0.1383 44 0.0034 317 0.1443 50 0.0031 323 0.1463 56 0.0026 329 0.1501 60 0.0022 333 0.1540 67
 0.0020 340 0.1563 71 0.0017 344 0.1604 76 0.00145 349 0.1646 85 0.0013 358 0.1676 90 0.0011 363 0.1724 96
 0.0010 369 0.1753 100 0.0009 373 0.1786

$$\frac{1}{\ln \frac{\eta}{\eta_0}}(T)$$

Having determined the parameters of the approximating straight line, we obtain a system of equations:

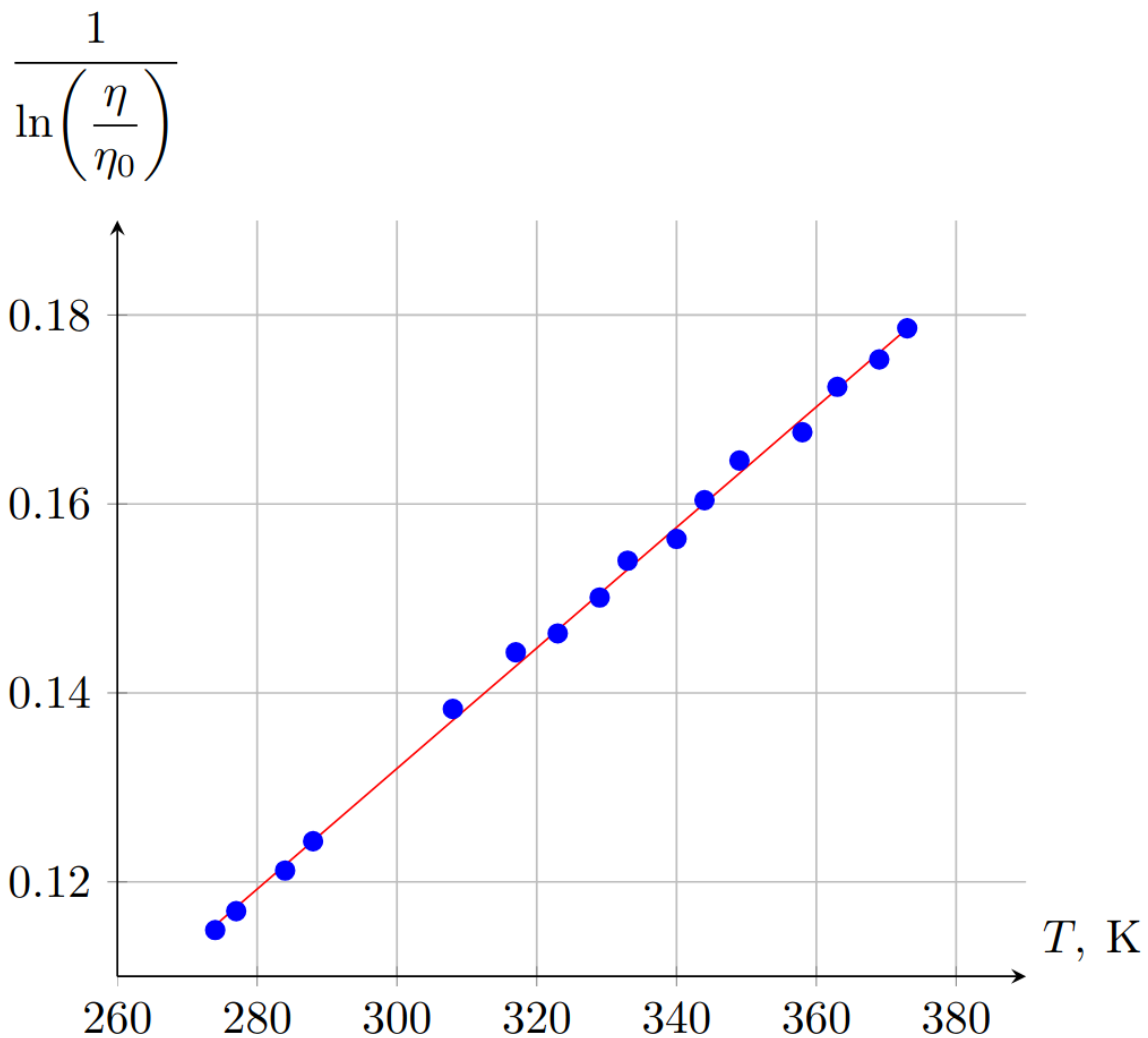
$$\begin{cases} -\frac{1}{D} = -0.059 \\ \frac{1}{DT_0} = 6.38 \cdot 10^{-4} K^{-1}, \end{cases}$$

which follows:

Answer:

$$T_0 = 92.5 \text{ K}$$

$$D = 16.9$$



C1 Determine the viscosity of the mixture η_n at the volume fraction of glycerol $c_g = 0.5$ and temperature $T = 25^\circ\text{C}$ using the dependence on the mole fraction of water.

0.30

The equation obtained for $\eta_n(n_w)$ is applicable over the entire range. Therefore, it is enough to convert c_g into n_w :

$$n_w = 1 - \frac{w_g \mu_w}{\mu_g - w_g(\mu_g - \mu_w)} = 1 - \frac{\frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} \mu_w}{\mu_g - \frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} (\mu_g - \mu_w)} \approx 0.80$$

and substitute it into the resulting equation. Thus we get:

Answer: $\eta_n = 6.0 \text{ mPa} \cdot \text{s}$

C2 Determine the viscosity of the mixture η_t under the same conditions using the temperature dependence.

0.30

Similar to the previous point, we substitute the value $T = 25^\circ\text{C} = 298\text{ K}$ into the equation obtained in part B, which follows:

Answer: $\eta_t = 6.8 \text{ mPa} \cdot \text{s}$

C3 Compare your results. If there are discrepancies, explain the reason for them.

0.20

Answer: Considering that the measurement range is large, the results can be considered coincident. The slight difference is most likely due to inaccurate temperature measurements, since the dependence of viscosity on temperature is very strong.

D1 Restore the missing values v_2 in the table. Use the linearized dependence graph $v_2(\eta)$ for this. Use the empty table columns on the answer sheet to convert the required values.

2.00

In the case of a hydrophilic capillary, the dependence of the volumetric flow rate of liquid on viscosity is represented by the equation:

$$Q = \pi r^2 v = \frac{\pi r^4 \Delta P}{8 \eta L} = \frac{\pi r^4 \rho g}{8 \eta}$$

, which implies the relation:

$$v = k \frac{\rho}{\eta},$$

where k is a constant value. The density of the mixture can be calculated as:

$$\rho = \rho_g c_g + \rho_w c_w = \rho_w + (\rho_g - \rho_w) c_g$$

Using the expression:

$$n_w = 1 - \frac{\frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} \mu_w}{\mu_g - \frac{\rho_g c_g}{(\rho_g - \rho_w)c_g + \rho_w} (\mu_g - \mu_w)},$$

we express c_g :

$$n_w(\mu_g \rho_w(1 - c_g) + \rho_g c_g \mu_w) = \mu_g \rho_w(1 - c_g)$$

$$c_g = \frac{\mu_g \rho_w(1 - n_w)}{\rho_g \mu_w + \rho_w \mu_g(1 - n_w)}$$

Then the expression for the density of the mixture becomes:

$$\rho = \rho_w + (\rho_g - \rho_w) \frac{\mu_g \rho_w(1 - n_w)}{\rho_g \mu_w + \rho_w \mu_g(1 - n_w)}$$

Using the results of part A, we numerically determine the values of n_w at $\eta < \xi$, then calculate the values of the densities of mixtures, then for each value velocity in the hydrophilic channel, we calculate the value of quantity $\frac{\rho}{\eta}$ and plot graph $v \left(\frac{\rho}{\eta} \right)$:

$\eta, \text{ мПа} \cdot \text{с}$	$v_1, \text{ мм/с}$	$v_2, \text{ мм/с}$	n_w	$\rho, \frac{\text{кг}}{\text{м}^3}$	$\frac{\rho}{\eta}, \frac{\text{с}}{\text{м}^2} \cdot 10^3$
0.95	4.25	649	1.000	1000	1050
1.18	4.86	602	0.978	1021	865
1.32	5.03	545	0.966	1032	782
1.92	8.03	408	0.926	1060	552
2.41	9.80	310	0.905	1072	445
3.95	15.2	285	0.847	1100	278
5.81	18.2	253	0.804	1115	192
13.2	24.3	185	0.707	1141	86.4
18.4	25.5	150	0.666	1150	62.5
24.8	29.3	117	0.627	1157	46.7
72.3	38.8	117	0.475	1177	16.3
78.0	40.3	117	0.463	1178	15.1
224	51.0	119	0.278	1194	5.33
515	55.3	120	0.110	1204	2.34
989	59.4	126	0.090	1260	0.900

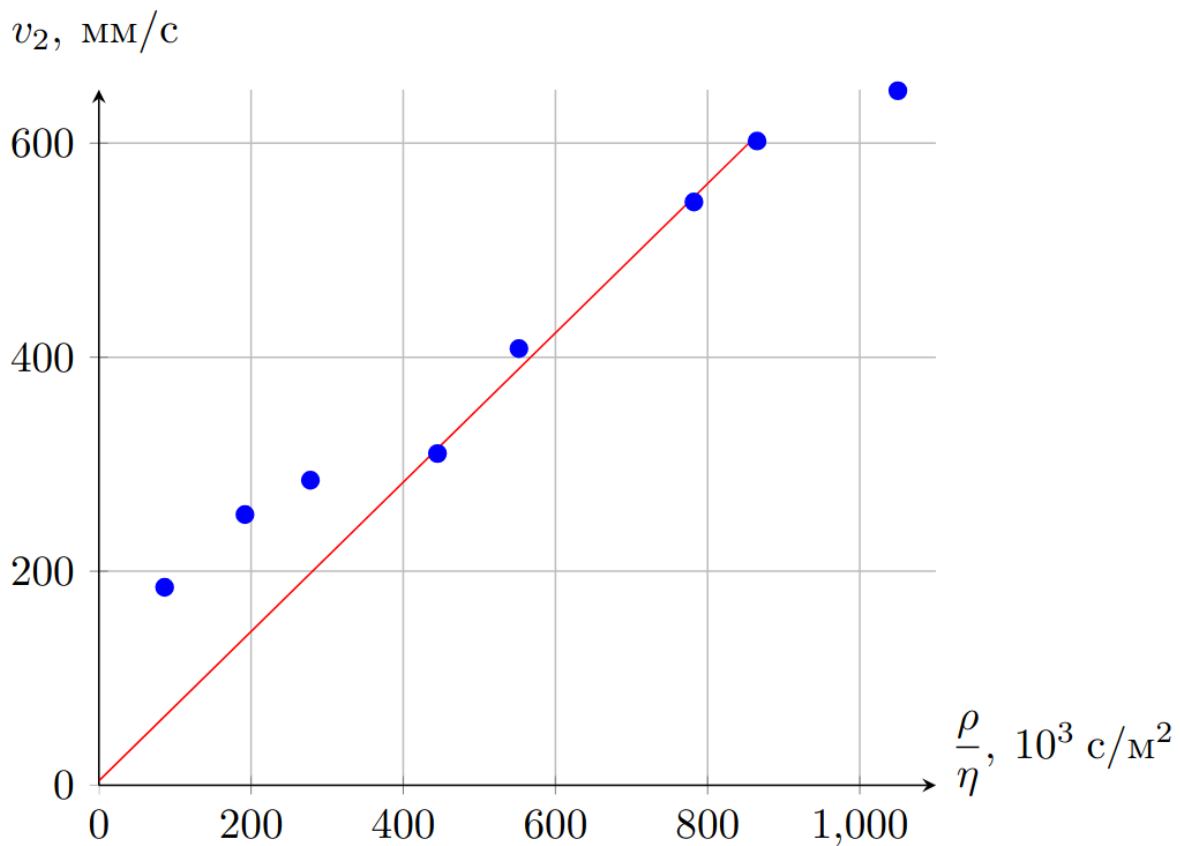
$v \left(\frac{\rho}{\eta} \right)$

The straight line must pass through (0, 0), therefore, drawing it along 4 points, we find k :

$$k = 0.674 \cdot 10^{-6} \frac{\text{м}^3}{\text{с}^2}$$

Now knowing k , we can restore the dependence $v_2(\eta)$:

$\eta, \text{ мПа} \cdot \text{с}$	$v_1, \text{ мм/с}$	$v_2, \text{ мм/с}$	n_w	$\rho, \frac{\text{кг}}{\text{м}^3}$	$\frac{\rho}{\eta}, \frac{\text{с}}{\text{м}^2} \cdot 10^3$
0.95	42.5	649	1.000	1000	1050
1.18	48.6	602	0.978	1021	865
1.32	50.3	545	0.966	1032	782
1.92	80.3	408	0.926	1060	552
2.41	98.0	310	0.905	1072	445
3.95	152	285	0.847	1100	278
5.81	182	253	0.804	1115	192
13.2	243	185	0.707	1141	86.4
18.4	255	150	0.666	1150	62.5
24.8	293	117	0.627	1157	46.7
72.3	388	110	0.475	1177	16.3
78.0	403	110	0.463	1178	15.1
224	510	119	0.278	1194	5.33
515	553	120	0.110	1204	2.34
989	594	126	0.090	1260	0.900



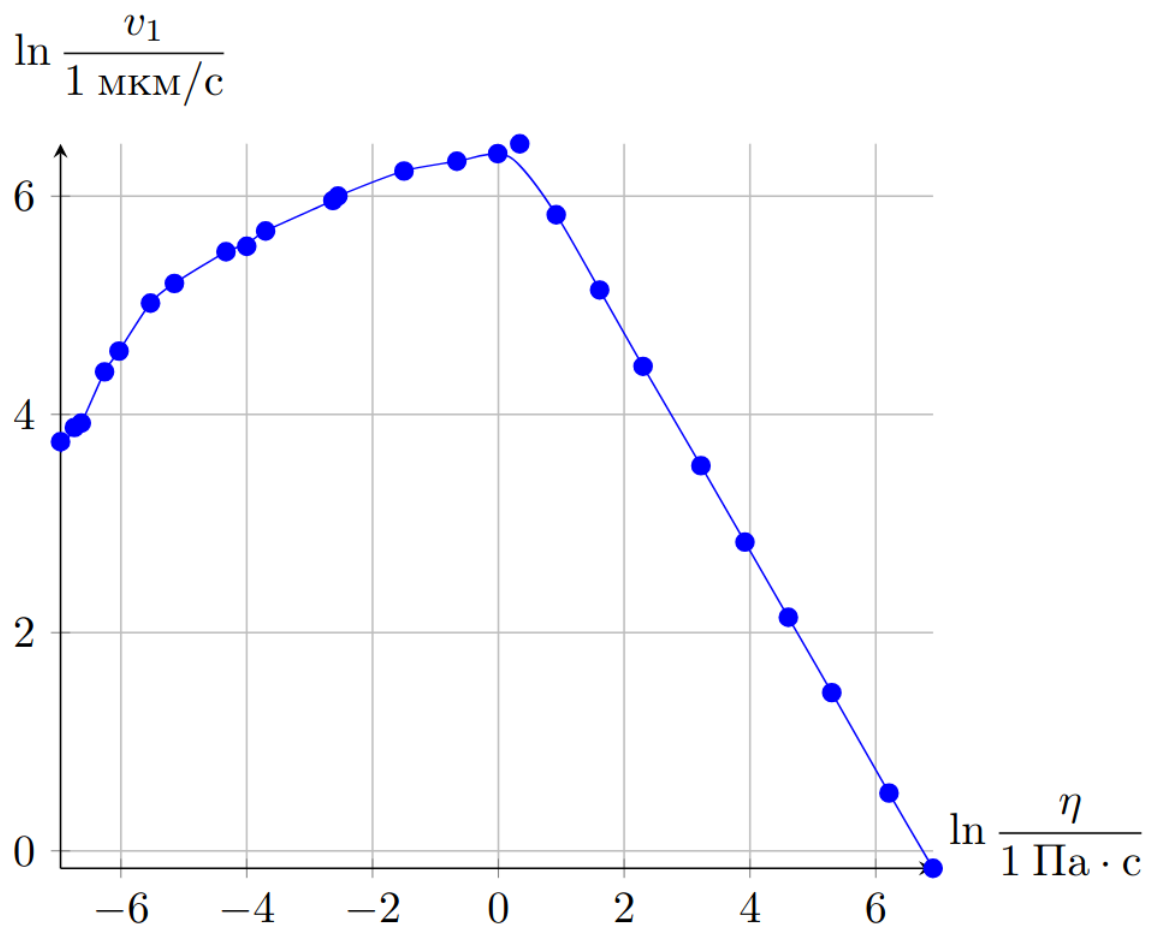
D2 Construct a graph of $v_1(\eta)$ in the range from $\eta_1 = 10^{-3} \Pi a \cdot c$ to $\eta_2 = 10^3 \Pi a \cdot c$ in the coordinates **1.50** that are most convenient for graphical analysis. All points plotted on the graph must be tabulated on the answer sheet.

At high viscosities, the velocities in hydrophobic and hydrophilic capillaries are equal, at $\eta = 1.400 \Pi a \cdot c$ $v_1 > v_2$, therefore we further assume that $v_1 = v_2$ at $\eta > \eta_{max} = 1.400 \Pi a \cdot c$. Let's enter the points into the table and construct a graph in logarithmic coordinates, since the values of the quantities under study vary by several orders of magnitude.

$\eta, \text{ mPa} \cdot \text{c}$	$v_1, \text{ мкм/с}$	$v_2, \text{ мм/с}$	n_w	$\rho, \frac{\text{kg}}{\text{м}^3}$	$\frac{c}{\rho/\eta}, \frac{\text{с}}{\text{м}^2}$	$\cdot 10^3 \ln \frac{v}{1 \text{ мм/с}}$	$\ln \frac{\eta}{1 \text{ Па} \cdot \text{с}}$	0.95	42.5	649	1.000	1000	1050															
3.75	-6.96	1.18	48.6	602	0.978	1021	865	3.88	-6.74	1.32	50.3	545	0.966	1032	782	3.92	-6.63	1.92	80.3	408	0.926							
1060	552	4.39	-6.26	2.41	98.0	310	0.905	1072	445	4.58	-6.03	3.95	152	285	0.847	1100	278	5.02	-5.53	5.81	182	253						
0.804	1115	192	5.20	-5.15	13.2	243	185	0.707	1141	86.4	5.49	-4.33	18.4	255	42.2	0.666	1150	62.5	5.54	-4.00	24.8							
293	31.5	0.627	1157	46.7	5.68	-3.70	72.3	388	11.0	0.475	1177	16.3	5.96	-2.63	78.0	403	10.2	0.463	1178	15.1	6.00							
-2.55	224	510	3.59	0.278	1194	5.33	6.23	-1.50	515	553	1.58	0.110	1204	2.34	6.32	-0.66	989	594	0.856	1260	1.27							
$\eta, \text{ Па} \cdot \text{с}$	$v_1, \text{ мкм/с}$	$v_2, \text{ мм/с}$	n_w	$\rho, \frac{\text{kg}}{\text{м}^3}$	$\frac{c}{\rho/\eta}, \frac{\text{с}}{\text{м}^2}$	$\ln \frac{v_1}{1 \text{ мкм/с}}$	$\ln \frac{\eta}{1 \text{ Па} \cdot \text{с}}$	2.5	340	1260	504	5.83	0.92	5.0	170	1260	252	5.14	1.61	10.0	84.9	1260	126	4.44	2.30	25.0	34.0	1260
50.4	3.53	3.22	50.0	17.0	1260	25.2	2.83	3.92	100	8.49	1260	12.6	2.14	4.61	200	4.25	1260	6.30	1.45	5.30	500	1.70						
1260	2.52	0.53	6.21	1000	0.85	1260	1.26	-0.16	6.91																			

Let's plot the obtained points on the graph:

$$\ln \frac{v_1}{1 \text{ мкм/с}} \left(\ln \frac{\eta}{1 \text{ Па} \cdot \text{с}} \right)$$



D3 Determine the value of the mixture viscosity η_{max} such that the value $v_2(\eta_{max})$ is maximum.

0.30

Answer:

$$\eta_{max} = 1.400 \text{ Pa} \cdot c$$